Ball & Plate System Description

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2. System Description

The ball and Plate setup consist of the following components (numbers refer to the numbers in the picture)

Linear motors (1,2,3)
Three linear motors of type P01-23x80/30x90 control the angle and the height of the plate. The three linear motors are positioned in a equilateral triangle.

Linear motor controls (4,5,6)
Each of the linear motors (LinMot B1100-VF) has its own controller. The controller outputs the current to the linear motor and processes the encoder signal form the motor for position measurement of the motor shaft.

Power supplies (8)
The top power supply powers the linear motor controls, the bottom one supplies power to the linear motors.

Plate (10)
The plate which is lifted and tilted by the linear motors

Camera (12)
The camera for the video stream to the Nvidia Jetson AGX.

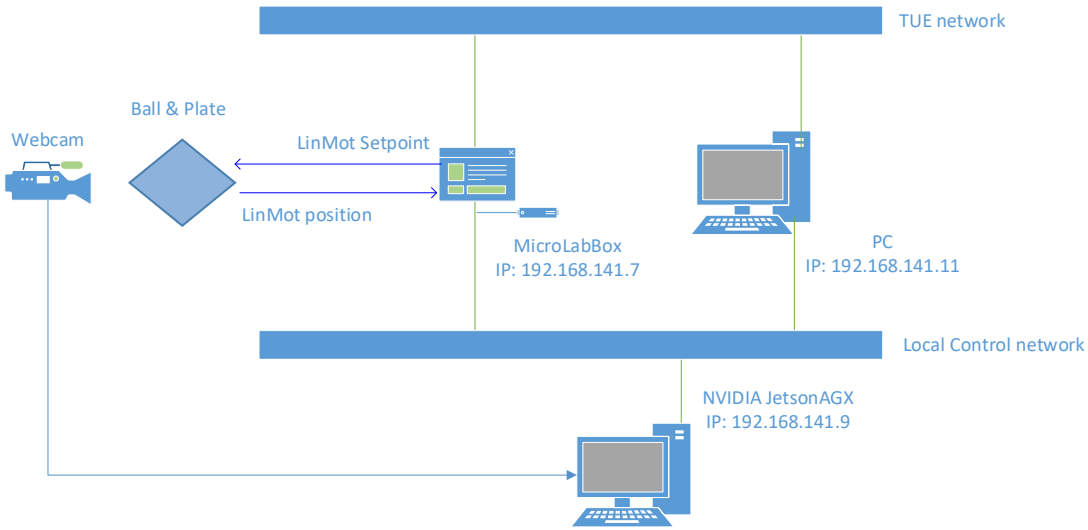
Nvidia Jetson AGX (9)
The stream from the camera is processes by the Nvidia Jetson AGX to determine the position of the ball on the plate.

Microlabbox (7)
The dSpace MicrolabBox runs the controller designed in Matlab/Simulink. It reads the various input signals (encoder signals from the motor-shafts, ball position) and outputs the reference signal for the linear motors.

Desktop PC (11)
Serves as the PC for controller development in Matlab/Simuink as well as for operating a controller in ControlDesk. Installed versions of Matlab/Simulink are R2019b and R2022b. This PC is also used for developing image-processing application on the Nvidia Jetson AGX using RTmaps-Remote-Studio.

3. Connecting the components

This section describes how the various components of the setup are connected, and which software components are used for data-communication between the various components. The various pieces of equipment are shown in the picture.



Important note:

The IP-addresses of the MicroLabBox on the Local Control Network is determined by the Simulink file that is loaded. The MicroLabBox operates as a switch for the Local Control Network, see description below. The Local Control Network does not exist until a (compiled) Simulink file is loaded into the MicroLabBox which sets the IP address of the MicroLabBox.

3.1 Connections

dSpace MicrolabBox - Desktop PC
This connection is for loading a compiled Matlab/Simulink application into the MicrolabBox.

Both the PC and the MicrolabBox are connected to the TU/e network. A Simulink application developed on the PC is compiled and then downloaded (ctrl B) over this network into the MicrolabBox where it is executed. ControlDesk on the PC is used to operate a running application which also communicates with the running application via the TU/e network.

Nvidia Jetson AGX – Microlabbox
This connection is for communicating the ball position, which is the result of image-processing on the Nvidia Jetson AGX.

The Simulink file also sets the port on which the application listens for information about the ball-position which is sent by the application running in the NVIDIA Jetson AGX. Ports and IP addresses have to match for this to work.

Nvidia Jetson AGX – Desktop PC
This connection over the Local Control Network is used for:

- Developing an image-processing application in RTMaps Studio for the NVIDIA Jetson AGX using RTMaps Studio.
- Starting and stopping an RTMaps application on the NVIDIA Jeston AGX.

The functionality of the application is to determine the ball-position on the plate using a camera.

Note: RTMaps applications on the NVIDIA Jetson AGX can run with and without a connection to RTMaps Studio on the PC, see the RTMaps section.

Linear motor Controllers to Microlabbox
The three linear motors receive a control signal from three DACs of the MicroLabBox. The encoders for shaft-position-measurements are connected to the encoder connector. Reading/Writing is via various software components in Matlab/Simulink, as described in Section 2.3.

3.1.1 Table of IP addresses and Ports

The following table lists the various IPaddresses and Ports of the equipment on the Local Control Network, as used in the various templates. Although they can be changed it is recommended not to.

Equipment	IP address	Port Type/Number	Function
MicroLabBox	192.168.141.7	UDP:49001	Listen for Ball Coordinates
NVIDIA Jetson AGX	192.168.141.9	UDP: any	Send Ball Coordinates
Desktop PC	192.168.141.11	TCP:10057	Development on NVIDIA Jetson AGX using Studio on the desktop PC

3.2 Simulink Software components for reading and writing variables.

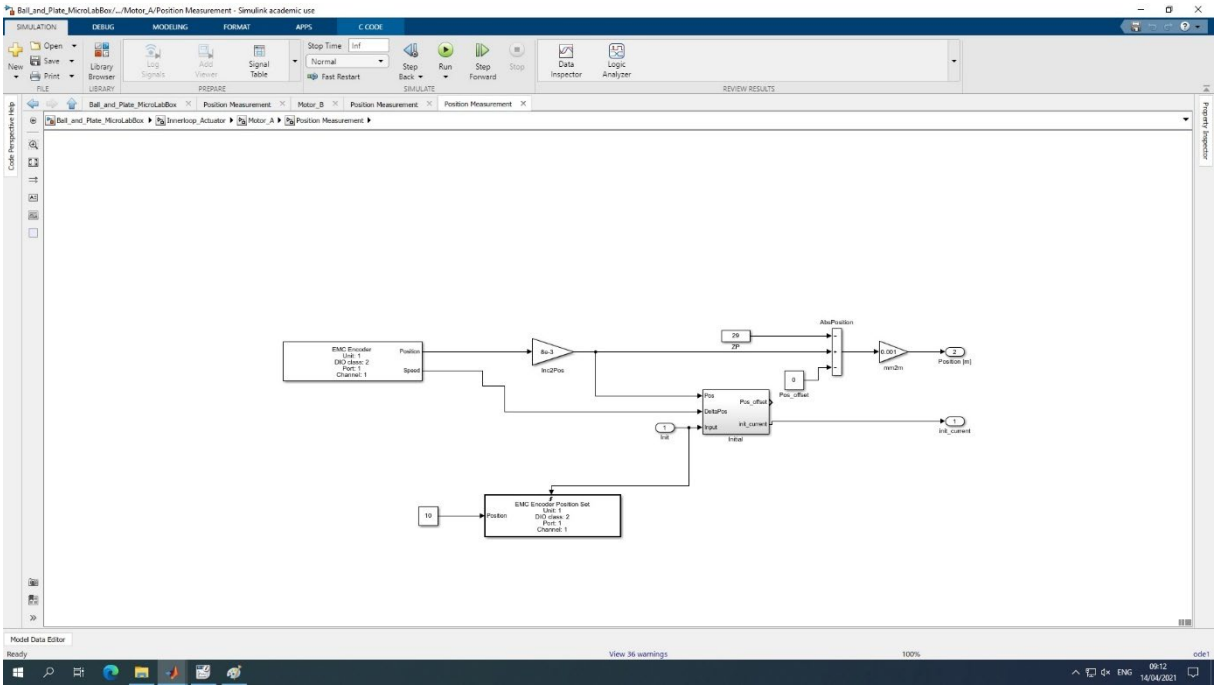
The MicroLabbox is known to Matlab/Simulink as RTI1202. Development of new Simulink models should start with the right template. The various software components discussed in this Section are made available in file sensors_and_actuators_ball_and_plate.slx.

Reading encoder signals of the linear motors

De encoder signals are read using the EMC_ENCODER_BL1 block. The encoder signal is converted into (mm), an offset (Zero Position) is subtracted and the result converted to (m).

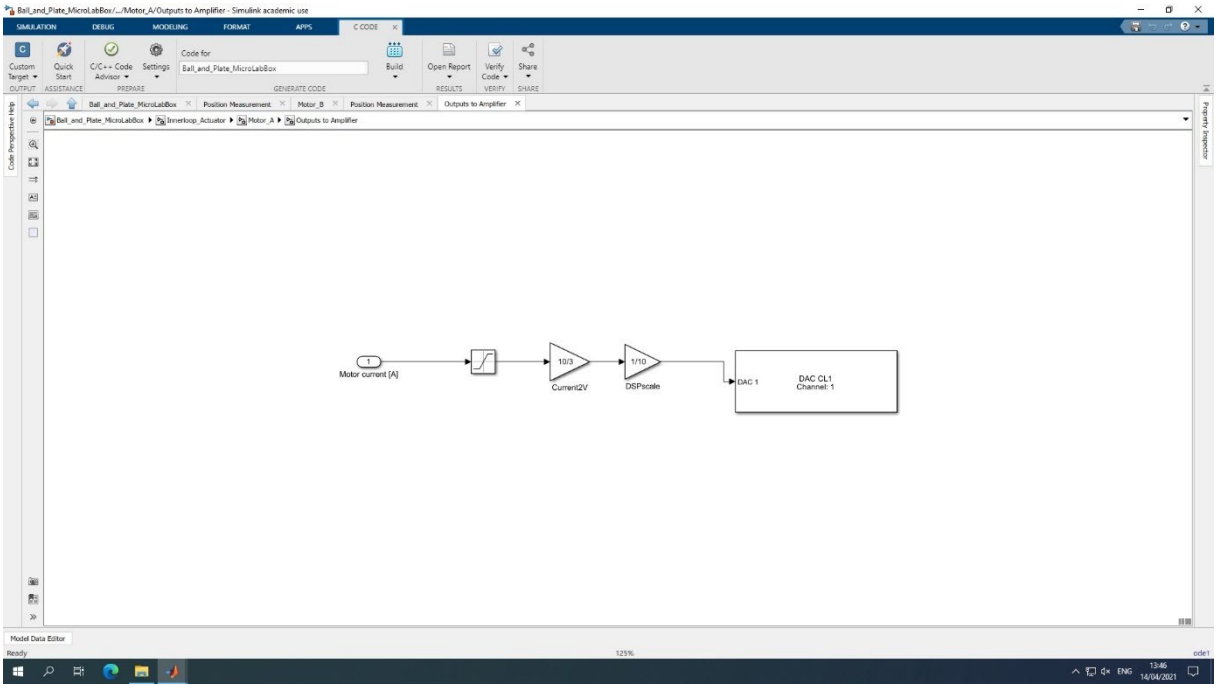
The plate should move to approximately the middle of the range as a starting position when switching on the controls. The full range is about 60(mm) so the plate has to move 30(mm) from its bottom position hence the Zero Position is about 30 (mm)

A separate block (EMC_ENCODER_POS_SET_BL1) resets the encoder to a preset-value to ensure the full encoder range is used. A good reset value is 10. The “init signal” should be set (and then reset) when the platform is in its rest position (unexcited motors).



Writing the control signal of the linear motors

The output signal of the current controller is limited and properly scaled before being written to a DAC channel on the MicrolabBox by the DAC-block. Each linear motor requires this scheme.



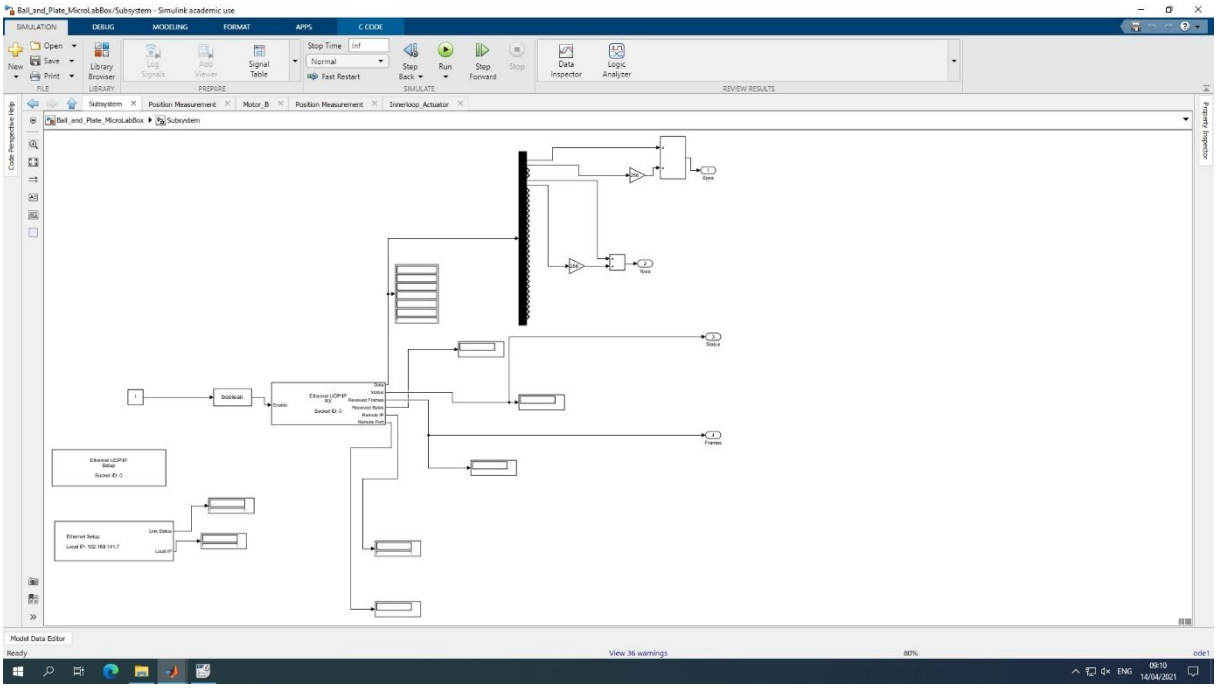
Note: Simulink Input -1…+1 results in a DAC output -10…+10 (V), hence the multiplier of 1/10.

Reading ball position on the plate

The Nvidia Jetson AGX generates a vector of 7 values of which the first two are the x- and y-coordinates in pixels of the centre of the blob-detection rectangle. This vector of seven values is sent to the MicrolabBox using the UDP-protocol. The received data-stream is 32 bytes long where the first two bytes represent the x-position and bytes 5 and 6 the y-position.

The Simulink application to read the values requires 3 blocks and a bus:

- Ethernet Setup: defines the IP address of MicrolabBox in the Local Control Network. The IPaddress to use is 192.168.141.7 with subnetmask 255.255.255.0.
- Ethernet UDP/IP setup: binds the UDP socket to the IP-address and port (49001).
- Ethernet UDP/IP RX: defines the data transfer parameters.
- Bus: exposes the data stream to other Simulink blocks.



4. Image Processing

An image processing application is developed in RTMaps for determining the ball-position from the camera-feed. This is described in the next section.

However, it is also possible to develop a custom application in Python on the NVIDIA Jetson AGX. Details are described in a separate section.

4.1 RTMaps

An image-processing application is available on the Nvidia Jetson AGX. The application sends the x- and y-coordinates in pixels over a Local Control Network using the UDP-protocol. It is developed using RTMaps for embedded systems.

Running the image-processing application assumes the Nvidia Jetson AGX is in the Local Control Network with IP address 192.168.141.9.

There are two ways to start the application:

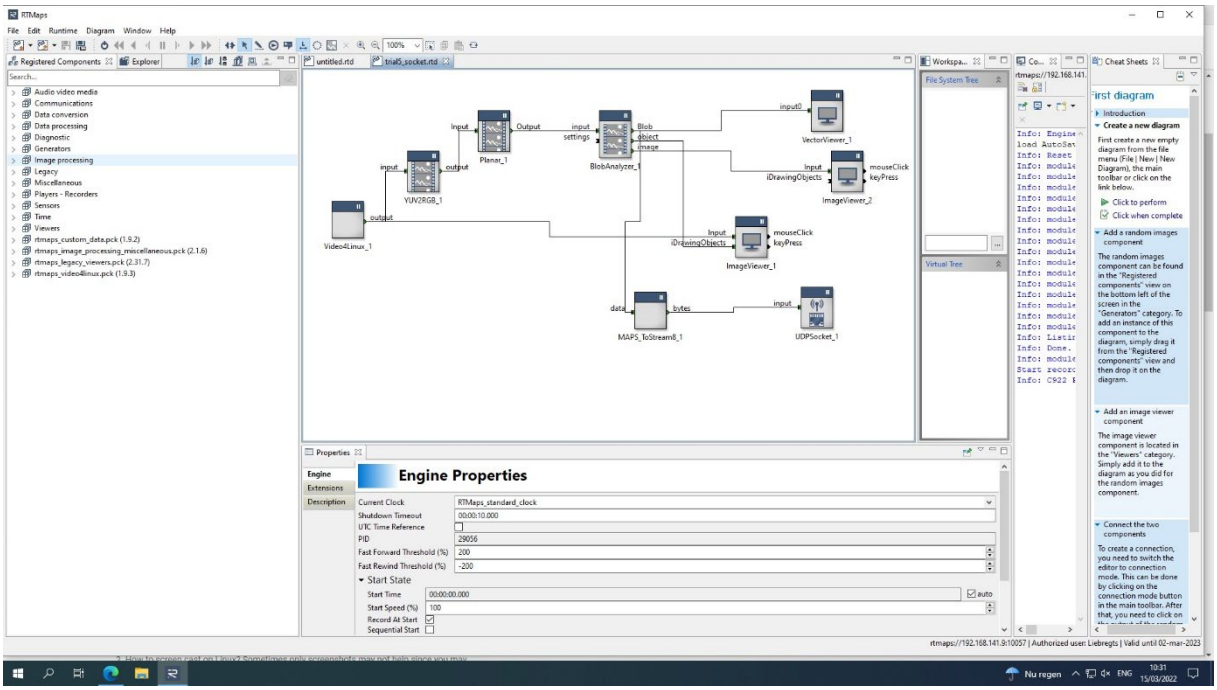
- From RTMaps Studio, which runs on the desktop PC.
- Start imageprocessing directly on the NVIDIA Jetson AGX.

4.1.1 Run Imageprocessing from RTMaps Studio

The following procedure starts the ball-detection algorithm from the desktop PC.

- On the NVIDIA Jetson AGX, start the RTMaps listener for connections to the desktop PC by the following command: /opt/rtmaps/bin/rtmaps_runtime –remote-studio-no-credentials.
- On the desktop PC start RTMaps Studio.
- In RTMaps Studio, in the Runtime menu, “Connect To Unmanaged Runtime”.
- Select or Add “rtmaps://192,168.141.9:10057”.
- On the File menu, open /home/ball_and_plate/Desktop/trial5_socket.rtd.

- The application is now showing and you can start and stop it by selecting the Run/Shutdown button in the menu bar.



4.1.2 Start Imagaprocessing directly on NVIDIA Jetson AGX

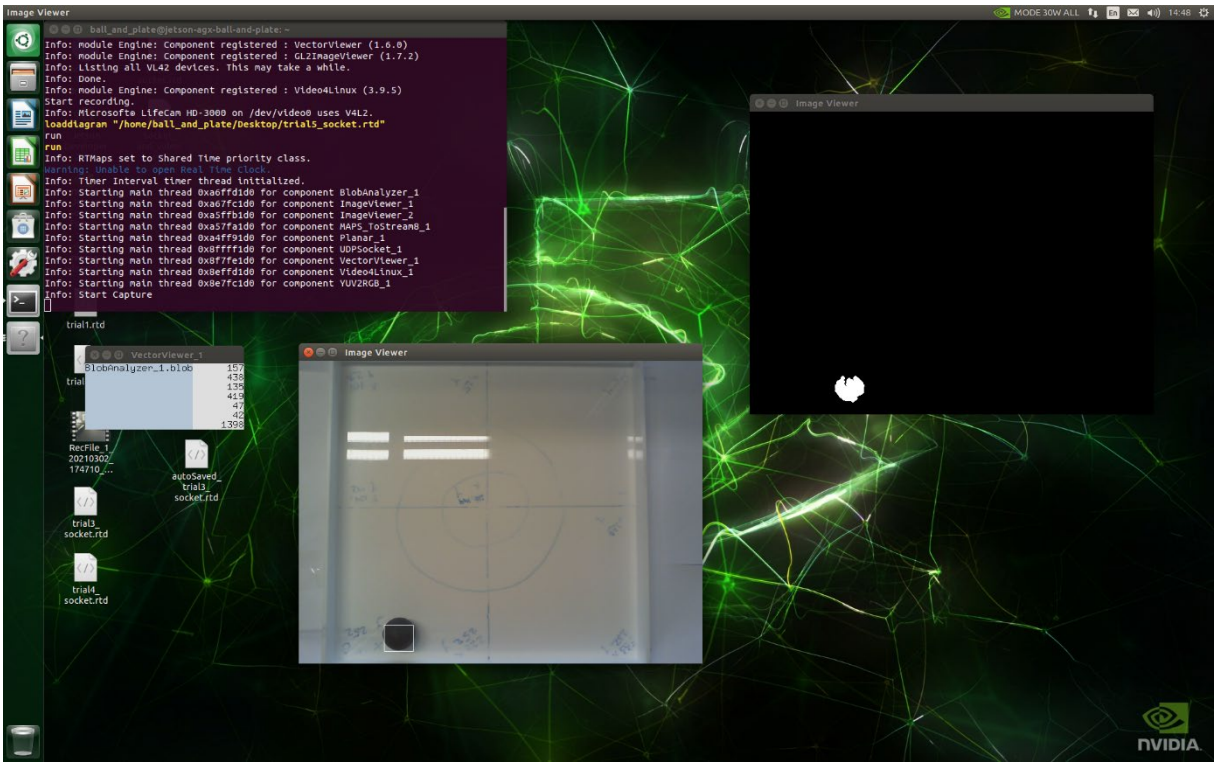
To start the imageprocessing application on NVIDIA Jetson AGX:

1. Open a terminal
2. To start RTmaps runtime enter: /opt/rtmaps/bin/rtmaps_runtime
3. When RTMaps is running enter: loaddiagram “/home/ball_and_plate/Desktop/ trial5_socket.rtd”
4. Run the application by entering: Run

Three windows will open:

- The raw video-feed with the blob-detection-box.
- A processed video-feed with a blob.
- The vector with coordinates (in pixels), sizes of the detection-box and size of the blob.

The camera-feed is a 640x480 image.



4.2 Image Processing in Python

The current setup allows for the development of a separate image-processing application in Python (3.6.9) This application runs on the NVIDIA Jetson AGX and is completely separate from RTMaps.

A file called test_socket_and_camera.py (in directory /home/ball_and_plate/Documents/OpenCV/opencv/samples/python) contains code for reading frames from the webcam using openCV, and writing the ball-position to the MicrolabBox using the socket module.

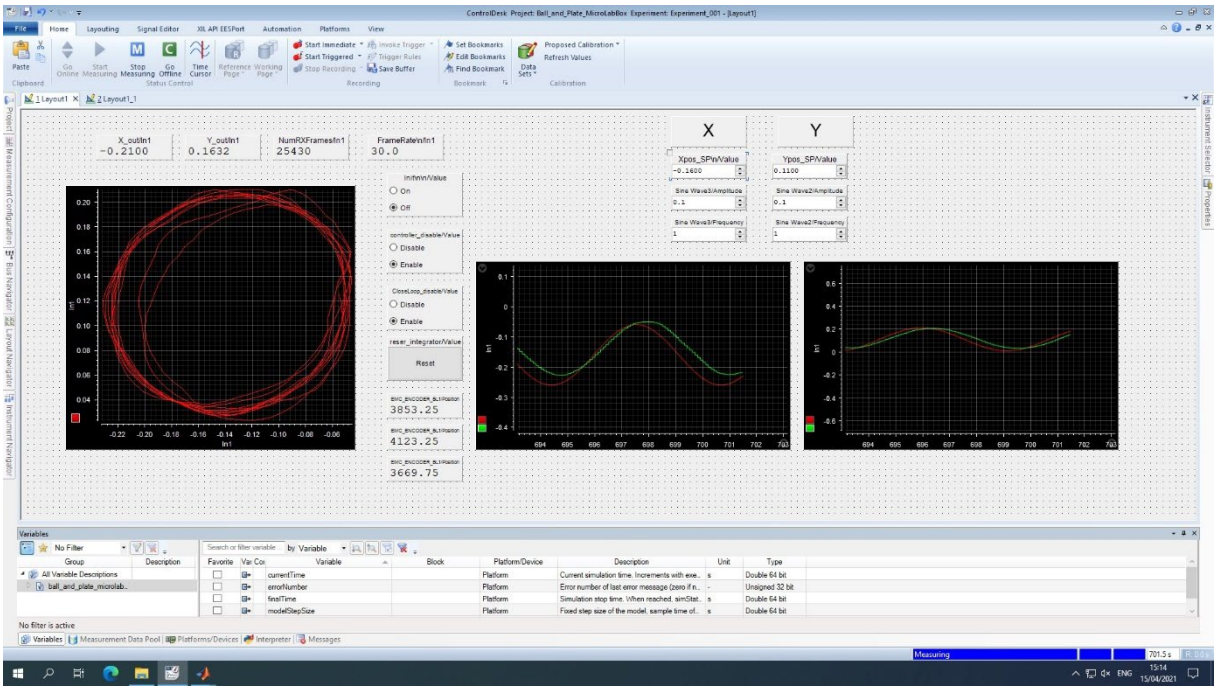
In a terminal, in the correct directory, the file can be run by entering “python3 test_socket_and_camera.py”.

The challenge is to develop an application to determine the ball-position from the webcam-frames with a sufficiently high sample frequency for the controller that runs in the MicroLabBox.

5. Operating the Ball & Plate setup

Visibility of values of variables in real-time is key to understanding the behavior of the setup. Controldesk will be used for this purpose. ControlDesk also has a data-collection tool (recorder) so that real-time data can be stored for further analysis in Matlab.

The user should develop their own dashboards in Controldesk projects/experiments. An example is given in the following screenshot:



The following videos are a good introduction to ControlDesk.

dSpace ControlDesk Measuring and Plotting Data:
https://www.youtube.com/watch?v=Cx9g75fVazU&t=15s

dSpace ControlDesk Instrumentation:
https://www.youtube.com/watch?v=6sa9LsG94xY&list=PLZNVoTBc8W3V68dJQ0Ll9yYPNh5cs6jW9&index=2

5.1 Typical Start sequence (with a controller)

For an existing controller, a typical start-sequence would be as follows:

1. Switch on the power supplies, switch on the individual motor controllers with the individual switches.
2. Run the appropriate .m files to define the motor and position controllers
3. Load Simulink file into the MicrolabBox (ctrl-b in Simulink)
4. Start the image-processing application on the Nvidia Jetson AGX
5. Initialize the motor encoders by switching the Init flag
6. Apply the position reference signal.
7. Start the controller by switching off the Controller-disable and CloseLoop_disable flags.
8. Change the position reference signal to test the controller

6. Troubleshooting

Red lights on Linear Motor controller

Toggle the on/off switch of the individual motor. If that does not solve the problem, switch off the power supply to the linear motor controller (24V) , wait 5 seconds and switch it back on.

Receiving no position data (ball-coordinates) while the image processing software is running.
This can happen when the controller was not running in the Microlabbox when the imageprocessing application was started. Stop and restart the application on the Nvidia Jetson AGX may resolve the issue.

7. System Data/Datasheets

There is a binder with datasheets and product information of the linear motors located next to the setup. The power supply parameters:

Description	Value
Power supply Linear Motor Controllers	24 (VDC)
Power Supply Linear Motors	50 (VDC)
Limit Linear Motor	-3 - +3 (A)

The plate is made of Perspex, with a density of approximately 1180 kg/m^3.